

Thermodynamics

Instructor

Research Assistant Buğra Kaplan

Chapter 5 – Problems

**Problem numbers correspond to
5th Edition**

5-7 Air enters a nozzle steadily at 2.21 kg/m^3 and 40 m/s and leaves at 0.762 kg/m^3 and 180 m/s . If the inlet area of the nozzle is 90 cm^2 , determine (a) the mass flow rate through the nozzle, and (b) the exit area of the nozzle. *Answers:* (a) 0.796 kg/s , (b) 58 cm^2

Assumptions Flow through the nozzle is steady.

Properties The density of air is given to be 2.21 kg/m^3 at the inlet, and 0.762 kg/m^3 at the exit.

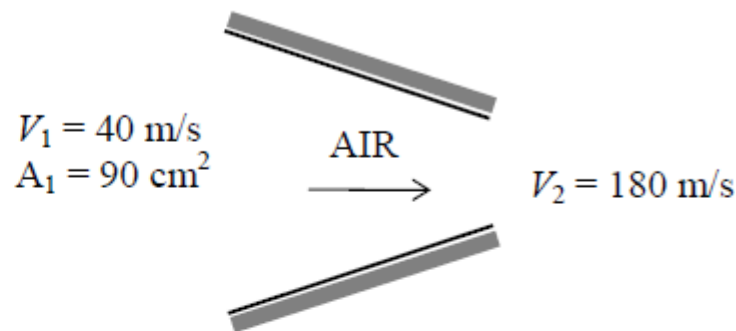
Analysis (a) The mass flow rate of air is determined from the inlet conditions to be

$$\dot{m} = \rho_1 A_1 V_1 = (2.21 \text{ kg/m}^3)(0.009 \text{ m}^2)(40 \text{ m/s}) = \mathbf{0.796 \text{ kg/s}}$$

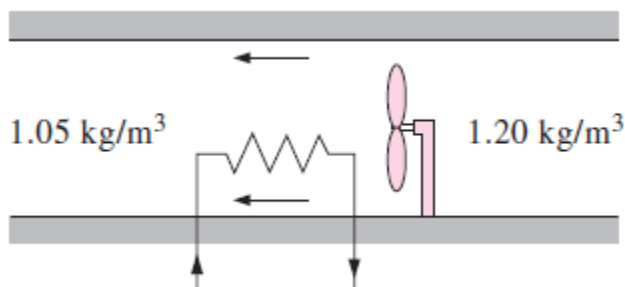
(b) There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$.

Then the exit area of the nozzle is determined to be

$$\dot{m} = \rho_2 A_2 V_2 \longrightarrow A_2 = \frac{\dot{m}}{\rho_2 V_2} = \frac{0.796 \text{ kg/s}}{(0.762 \text{ kg/m}^3)(180 \text{ m/s})} = 0.0058 \text{ m}^2 = \mathbf{58 \text{ cm}^2}$$



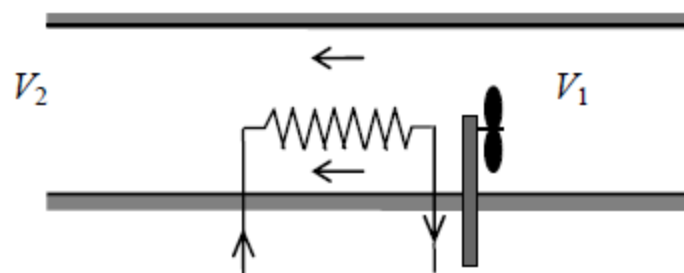
5-8 A hair dryer is basically a duct of constant diameter in which a few layers of electric resistors are placed. A small fan pulls the air in and forces it through the resistors where it is heated. If the density of air is 1.20 kg/m^3 at the inlet and 1.05 kg/m^3 at the exit, determine the percent increase in the velocity of air as it flows through the dryer.



Properties The density of air is given to be 1.20 kg/m^3 at the inlet, and 1.05 kg/m^3 at the exit.

Analysis There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. Then,

$$\begin{aligned}\dot{m}_1 &= \dot{m}_2 \\ \rho_1 A V_1 &= \rho_2 A V_2 \\ \frac{V_2}{V_1} &= \frac{\rho_1}{\rho_2} = \frac{1.20 \text{ kg/m}^3}{1.05 \text{ kg/m}^3} = 1.14 \quad (\text{or, an increase of } \mathbf{14\%})\end{aligned}$$



Therefore, the air velocity increases 14% as it flows through the hair drier.

5–10 A 1-m^3 rigid tank initially contains air whose density is 1.18 kg/m^3 . The tank is connected to a high-pressure supply line through a valve. The valve is opened, and air is allowed to enter the tank until the density in the tank rises to 7.20 kg/m^3 . Determine the mass of air that has entered the tank.

Answer: 6.02 kg

Properties The density of air is given to be 1.18 kg/m^3 at the beginning, and 7.20 kg/m^3 at the end.

Analysis We take the tank as the system, which is a control volume since mass crosses the boundary. The mass balance for this system can be expressed as

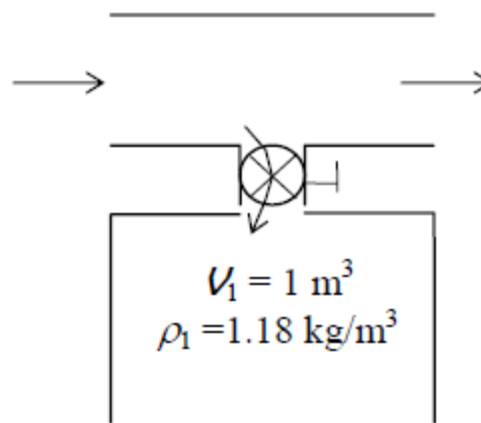
Mass balance:

$$m_{in} - m_{out} = \Delta m_{\text{system}} \rightarrow m_i = m_2 - m_1 = \rho_2 V - \rho_1 V$$

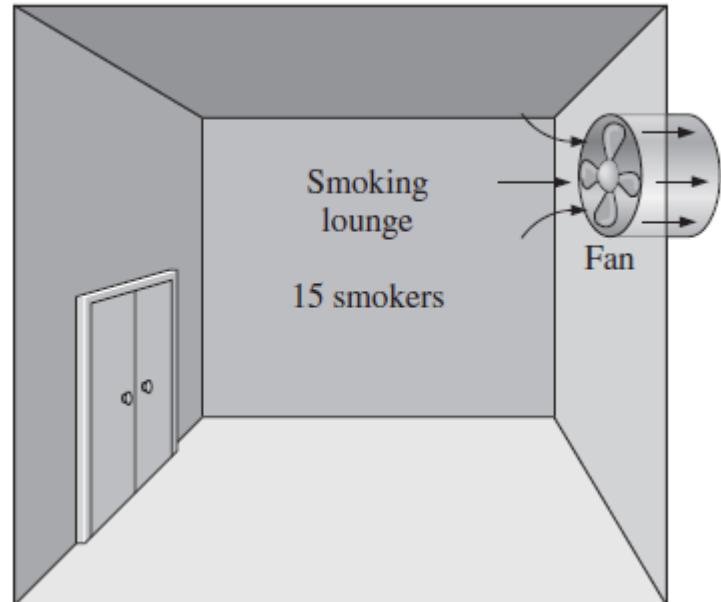
Substituting,

$$m_i = (\rho_2 - \rho_1)V = [(7.20 - 1.18)\text{ kg/m}^3](1\text{ m}^3) = \mathbf{6.02\text{ kg}}$$

Therefore, 6.02 kg of mass entered the tank.

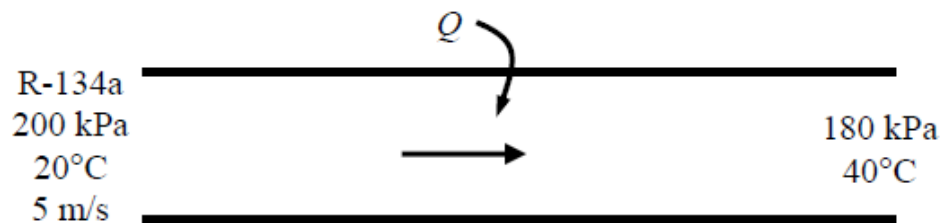


5-13 A smoking lounge is to accommodate 15 heavy smokers. The minimum fresh air requirement for smoking lounges is specified to be 30 L/s per person (ASHRAE, Standard 62, 1989). Determine the minimum required flow rate of fresh air that needs to be supplied to the lounge, and the diameter of the duct if the air velocity is not to exceed 8 m/s.



5-14 The minimum fresh air requirement of a residential building is specified to be 0.35 air change per hour (ASHRAE, Standard 62, 1989). That is, 35 percent of the entire air contained in a residence should be replaced by fresh outdoor air every hour. If the ventilation requirement of a 2.7-m-high, 200-m² residence is to be met entirely by a fan, determine the flow capacity in L/min of the fan that needs to be installed. Also determine the diameter of the duct if the air velocity is not to exceed 6 m/s.

5-16 Refrigerant-134a enters a 28-cm diameter pipe steadily at 200 kPa and 20°C with a velocity of 5 m/s. The refrigerant gains heat as it flows and leaves the pipe at 180 kPa and 40°C. Determine (a) the volume flow rate of the refrigerant at the inlet, (b) the mass flow rate of the refrigerant, and (c) the velocity and volume flow rate at the exit.



Properties The specific volumes of R-134a at the inlet and exit are (Table A-13)

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa} \\ T_1 = 20^\circ\text{C} \end{array} \right\} \nu_1 = 0.1142 \text{ m}^3/\text{kg} \qquad \left. \begin{array}{l} P_1 = 180 \text{ kPa} \\ T_1 = 40^\circ\text{C} \end{array} \right\} \nu_2 = 0.1374 \text{ m}^3/\text{kg}$$

Analysis (a) (b) The volume flow rate at the inlet and the mass flow rate are

$$\dot{V}_1 = A_c V_1 = \frac{\pi D^2}{4} V_1 = \frac{\pi (0.28 \text{ m})^2}{4} (5 \text{ m/s}) = \mathbf{0.3079 \text{ m}^3/\text{s}}$$

$$\dot{m} = \frac{1}{\nu_1} A_c V_1 = \frac{1}{\nu_1} \frac{\pi D^2}{4} V_1 = \frac{1}{0.1142 \text{ m}^3/\text{kg}} \frac{\pi (0.28 \text{ m})^2}{4} (5 \text{ m/s}) = \mathbf{2.696 \text{ kg/s}}$$

(c) Noting that mass flow rate is constant, the volume flow rate and the velocity at the exit of the pipe are determined from

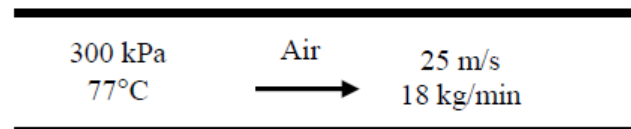
$$\dot{V}_2 = \dot{m} \nu_2 = (2.696 \text{ kg/s})(0.1374 \text{ m}^3/\text{kg}) = \mathbf{0.3705 \text{ m}^3/\text{s}}$$

$$V_2 = \frac{\dot{V}_2}{A_c} = \frac{0.3705 \text{ m}^3/\text{s}}{\frac{\pi (0.28 \text{ m})^2}{4}} = \mathbf{6.02 \text{ m/s}}$$

5-24 Air flows steadily in a pipe at 300 kPa, 77°C, and 25 m/s at a rate of 18 kg/min. Determine (a) the diameter of the pipe, (b) the rate of flow energy, (c) the rate of energy transport by mass, and (d) also determine the error involved in part (c) if the kinetic energy is neglected.

Properties The properties of air are $R = 0.287 \text{ kJ/kg}\cdot\text{K}$ and $c_p = 1.008 \text{ kJ/kg}\cdot\text{K}$ (at 350 K from Table A-2b)

Analysis (a) The diameter is determined as follows



$$\nu = \frac{RT}{P} = \frac{(0.287 \text{ kJ/kg}\cdot\text{K})(77 + 273 \text{ K})}{(300 \text{ kPa})} = 0.3349 \text{ m}^3/\text{kg}$$

$$A = \frac{\dot{m}\nu}{V} = \frac{(18 / 60 \text{ kg/s})(0.3349 \text{ m}^3/\text{kg})}{25 \text{ m/s}} = 0.004018 \text{ m}^2$$

$$D = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(0.004018 \text{ m}^2)}{\pi}} = \mathbf{0.0715 \text{ m}}$$

(b) The rate of flow energy is determined from

$$\dot{W}_{\text{flow}} = \dot{m}P\nu = (18 / 60 \text{ kg/s})(300 \text{ kPa})(0.3349 \text{ m}^3/\text{kg}) = \mathbf{30.14 \text{ kW}}$$

(c) The rate of energy transport by mass is

$$\begin{aligned} \dot{E}_{\text{mass}} &= \dot{m}(h + ke) = \dot{m}\left(c_p T + \frac{1}{2}V^2\right) \\ &= (18/60 \text{ kg/s})\left[(1.008 \text{ kJ/kg}\cdot\text{K})(77 + 273 \text{ K}) + \frac{1}{2}(25 \text{ m/s})^2\left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2}\right)\right] \\ &= \mathbf{105.94 \text{ kW}} \end{aligned}$$

(d) If we neglect kinetic energy in the calculation of energy transport by mass

$$\dot{E}_{\text{mass}} = \dot{m}h = \dot{m}c_p T = (18/60 \text{ kg/s})(1.005 \text{ kJ/kg}\cdot\text{K})(77 + 273 \text{ K}) = 105.84 \text{ kW}$$

Therefore, the error involved if neglect the kinetic energy is only **0.09%**.

Steady-Flow Energy Balance: Nozzles and Diffusers

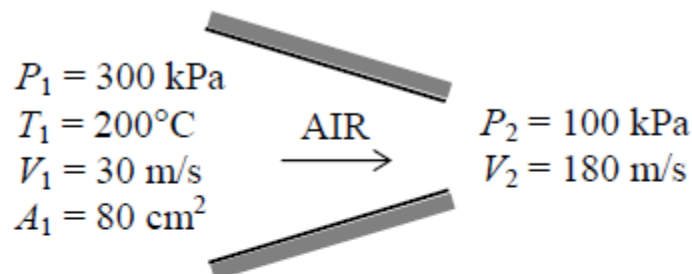
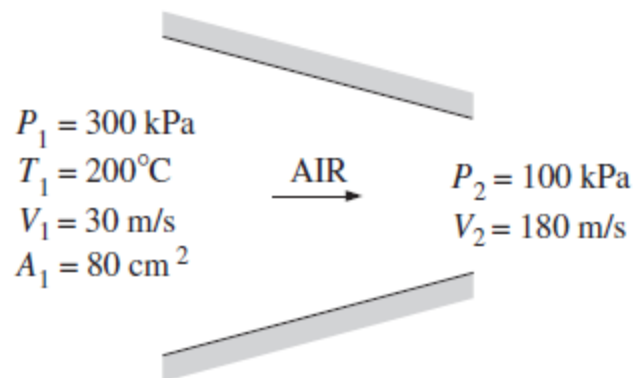
5-30 Air enters an adiabatic nozzle steadily at 300 kPa, 200°C, and 30 m/s and leaves at 100 kPa and 180 m/s. The inlet area of the nozzle is 80 cm². Determine (a) the mass flow rate through the nozzle, (b) the exit temperature of the air, and (c) the exit area of the nozzle. *Answers: (a) 0.5304 kg/s, (b) 184.6°C, (c) 38.7 cm²*

Properties The gas constant of air is 0.287 kPa·m³/kg·K (Table A-1). The specific heat of air at the anticipated average temperature of 450 K is $c_p = 1.02$ kJ/kg·°C (Table A-2).

Analysis (a) There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. Using the ideal gas relation, the specific volume and the mass flow rate of air are determined to be

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(473 \text{ K})}{300 \text{ kPa}} = 0.4525 \text{ m}^3/\text{kg}$$

$$\dot{m} = \frac{1}{v_1} A_1 V_1 = \frac{1}{0.4525 \text{ m}^3/\text{kg}} (0.008 \text{ m}^2)(30 \text{ m/s}) = \mathbf{0.5304 \text{ kg/s}}$$



(b) We take nozzle as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}(h_1 + V_1^2 / 2) = \dot{m}(h_2 + V_2^2 / 2) \quad (\text{since } \dot{Q} \cong \dot{W} \cong \Delta p e \cong 0)$$

$$0 = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} \longrightarrow 0 = c_{p,ave}(T_2 - T_1) + \frac{V_2^2 - V_1^2}{2}$$

$$\text{Substituting,} \quad 0 = (1.02 \text{ kJ/kg} \cdot \text{K})(T_2 - 200^\circ \text{C}) + \frac{(180 \text{ m/s})^2 - (30 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right)$$

$$\text{It yields} \quad T_2 = \mathbf{184.6^\circ \text{C}}$$

(c) The specific volume of air at the nozzle exit is

$$\nu_2 = \frac{RT_2}{P_2} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(184.6 + 273 \text{ K})}{100 \text{ kPa}} = 1.313 \text{ m}^3/\text{kg}$$

$$\dot{m} = \frac{1}{\nu_2} A_2 V_2 \longrightarrow 0.5304 \text{ kg/s} = \frac{1}{1.313 \text{ m}^3/\text{kg}} A_2 (180 \text{ m/s}) \rightarrow A_2 = 0.00387 \text{ m}^2 = \mathbf{38.7 \text{ cm}^2}$$

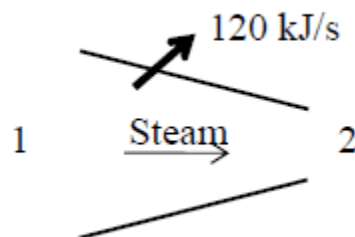
5-32 Steam at 5 MPa and 400°C enters a nozzle steadily with a velocity of 80 m/s, and it leaves at 2 MPa and 300°C. The inlet area of the nozzle is 50 cm², and heat is being lost at a rate of 120 kJ/s. Determine (a) the mass flow rate of the steam, (b) the exit velocity of the steam, and (c) the exit area of the nozzle.

Properties From the steam tables (Table A-6)

$$\left. \begin{array}{l} P_1 = 5 \text{ MPa} \\ T_1 = 400^\circ\text{C} \end{array} \right\} \begin{array}{l} \nu_1 = 0.057838 \text{ m}^3/\text{kg} \\ h_1 = 3196.7 \text{ kJ/kg} \end{array}$$

and

$$\left. \begin{array}{l} P_2 = 2 \text{ MPa} \\ T_2 = 300^\circ\text{C} \end{array} \right\} \begin{array}{l} \nu_2 = 0.12551 \text{ m}^3/\text{kg} \\ h_2 = 3024.2 \text{ kJ/kg} \end{array}$$



Analysis (a) There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. The mass flow rate of steam is

$$\dot{m} = \frac{1}{\nu_1} V_1 A_1 = \frac{1}{0.057838 \text{ m}^3/\text{kg}} (80 \text{ m/s})(50 \times 10^{-4} \text{ m}^2) = \mathbf{6.92 \text{ kg/s}}$$



(b) We take nozzle as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\dot{\Delta E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}(h_1 + V_1^2 / 2) = \dot{Q}_{\text{out}} + \dot{m}(h_2 + V_2^2 / 2) \quad (\text{since } \dot{W} \cong \Delta p e \cong 0)$$

$$-\dot{Q}_{\text{out}} = \dot{m} \left(h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} \right)$$

Substituting, the exit velocity of the steam is determined to be

$$-120 \text{ kJ/s} = (6.916 \text{ kg/s}) \left(3024.2 - 3196.7 + \frac{V_2^2 - (80 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) \right)$$

It yields $V_2 = \mathbf{562.7 \text{ m/s}}$

(c) The exit area of the nozzle is determined from

$$\dot{m} = \frac{1}{v_2} V_2 A_2 \longrightarrow A_2 = \frac{\dot{m} v_2}{V_2} = \frac{(6.916 \text{ kg/s})(0.12551 \text{ m}^3/\text{kg})}{562.7 \text{ m/s}} = \mathbf{15.42 \times 10^{-4} \text{ m}^2}$$

5-34

Steam at 3 MPa and 400°C enters an adiabatic nozzle steadily with a velocity of 40 m/s and leaves at 2.5 MPa and 300 m/s. Determine (a) the exit temperature and (b) the ratio of the inlet to exit area A_1/A_2 .

5-36

Air at 80 kPa and 127°C enters an adiabatic diffuser steadily at a rate of 6000 kg/h and leaves at 100 kPa. The velocity of the airstream is decreased from 230 to 30 m/s as it passes through the diffuser. Find (a) the exit temperature of the air and (b) the exit area of the diffuser.

5-38 Carbon dioxide enters an adiabatic nozzle steadily at 1 MPa and 500°C with a mass flow rate of 6000 kg/h and leaves at 100 kPa and 450 m/s. The inlet area of the nozzle is 40 cm². Determine (a) the inlet velocity and (b) the exit temperature. *Answers: (a) 60.8 m/s, (b) 685.8 K*

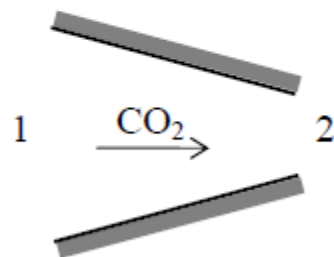
Properties The gas constant and molar mass of CO₂ are 0.1889 kPa·m³/kg·K and 44 kg/kmol (Table A-1). The enthalpy of CO₂ at 500°C is $\bar{h}_1 = 30,797$ kJ/kmol (Table A-20).

Analysis (a) There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. Using the ideal gas relation, the specific volume is determined to be

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.1889 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(773 \text{ K})}{1000 \text{ kPa}} = 0.146 \text{ m}^3/\text{kg}$$

Thus,

$$\dot{m} = \frac{1}{v_1} A_1 V_1 \longrightarrow V_1 = \frac{\dot{m} v_1}{A_1} = \frac{(6000/3600 \text{ kg/s})(0.146 \text{ m}^3/\text{kg})}{40 \times 10^{-4} \text{ m}^2} = 60.8 \text{ m/s}$$



(b) We take nozzle as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \overset{\text{No (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}(h_1 + V_1^2 / 2) = \dot{m}(h_2 + V_2^2 / 2) \quad (\text{since } \dot{Q} \cong \dot{W} \cong \Delta p_e \cong 0)$$

$$0 = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2}$$

Substituting,

$$\begin{aligned} \bar{h}_2 &= \bar{h}_1 - \frac{V_2^2 - V_1^2}{2} M \\ &= 30,797 \text{ kJ/kmol} - \frac{(450 \text{ m/s})^2 - (60.8 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) (44 \text{ kg/kmol}) \\ &= 26,423 \text{ kJ/kmol} \end{aligned}$$

Then the exit temperature of CO₂ from Table A-20 is obtained to be $T_2 = 685.8 \text{ K}$

5-39 Refrigerant-134a at 700 kPa and 120°C enters an adiabatic nozzle steadily with a velocity of 20 m/s and leaves at 400 kPa and 30°C. Determine (a) the exit velocity and (b) the ratio of the inlet to exit area A_1/A_2 .

Properties From the refrigerant tables (Table A-13)

$$\left. \begin{array}{l} P_1 = 700 \text{ kPa} \\ T_1 = 120^\circ\text{C} \end{array} \right\} \begin{array}{l} v_1 = 0.043358 \text{ m}^3/\text{kg} \\ h_1 = 358.90 \text{ kJ/kg} \end{array}$$

and

$$\left. \begin{array}{l} P_2 = 400 \text{ kPa} \\ T_2 = 30^\circ\text{C} \end{array} \right\} \begin{array}{l} v_2 = 0.056796 \text{ m}^3/\text{kg} \\ h_2 = 275.07 \text{ kJ/kg} \end{array}$$



Analysis (a) There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. We take nozzle as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\dot{\Delta E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}(h_1 + V_1^2/2) = \dot{m}(h_2 + V_2^2/2) \quad (\text{since } \dot{Q} \equiv \dot{W} \equiv \Delta p_e \equiv 0)$$

$$0 = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2}$$

Substituting,

$$0 = (275.07 - 358.90) \text{ kJ/kg} + \frac{V_2^2 - (20 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right)$$

It yields $V_2 = 409.9 \text{ m/s}$

(b) The ratio of the inlet to exit area is determined from the conservation of mass relation,

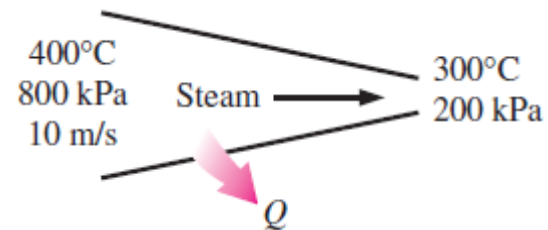
$$\frac{1}{v_2} A_2 V_2 = \frac{1}{v_1} A_1 V_1 \longrightarrow \frac{A_1}{A_2} = \frac{v_1}{v_2} \frac{V_2}{V_1} = \frac{(0.043358 \text{ m}^3/\text{kg})(409.9 \text{ m/s})}{(0.056796 \text{ m}^3/\text{kg})(20 \text{ m/s})} = \mathbf{15.65}$$

5-41 Nitrogen gas at 60 kPa and 7°C enters an adiabatic diffuser steadily with a velocity of 200 m/s and leaves at 85 kPa and 22°C. Determine (a) the exit velocity of the nitrogen and (b) the ratio of the inlet to exit area A_1/A_2 .

5-43 Refrigerant-134a enters a diffuser steadily as saturated vapor at 800 kPa with a velocity of 120 m/s, and it leaves at 900 kPa and 40°C. The refrigerant is gaining heat at a rate of 2 kJ/s as it passes through the diffuser. If the exit area is 80 percent greater than the inlet area, determine (a) the exit velocity and (b) the mass flow rate of the refrigerant.

Answers: (a) 60.8 m/s, (b) 1.308 kg/s

5-44 Steam enters a nozzle at 400°C and 800 kPa with a velocity of 10 m/s, and leaves at 300°C and 200 kPa while losing heat at a rate of 25 kW. For an inlet area of 800 cm², determine the velocity and the volume flow rate of the steam at the nozzle exit. *Answers: 606 m/s, 2.74 m³/s*



5-49 Steam flows steadily through an adiabatic turbine. The inlet conditions of the steam are 10 MPa, 450°C, and 80 m/s, and the exit conditions are 10 kPa, 92 percent quality, and 50 m/s. The mass flow rate of the steam is 12 kg/s. Determine (a) the change in kinetic energy, (b) the power output, and (c) the turbine inlet area. *Answers: (a) -1.95 kJ/kg, (b) 10.2 MW, (c) 0.00447 m²*

Properties From the steam tables (Tables A-4 through 6)

$$\left. \begin{array}{l} P_1 = 10 \text{ MPa} \\ T_1 = 450^\circ\text{C} \end{array} \right\} \begin{array}{l} v_1 = 0.029782 \text{ m}^3/\text{kg} \\ h_1 = 3242.4 \text{ kJ/kg} \end{array}$$

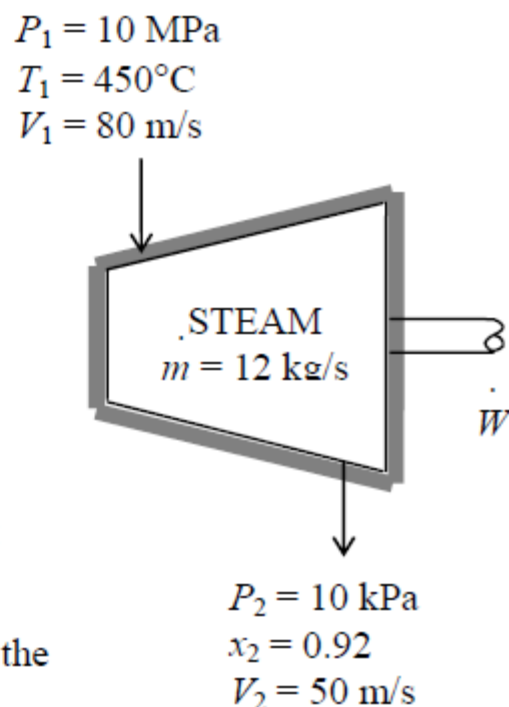
and

$$\left. \begin{array}{l} P_2 = 10 \text{ kPa} \\ x_2 = 0.92 \end{array} \right\} h_2 = h_f + x_2 h_{fg} = 191.81 + 0.92 \times 2392.1 = 2392.5 \text{ kJ/kg}$$

Analysis (a) The change in kinetic energy is determined from

$$\Delta ke = \frac{V_2^2 - V_1^2}{2} = \frac{(50 \text{ m/s})^2 - (80 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = -1.95 \text{ kJ/kg}$$

(b) There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. We take the turbine as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as



$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\dot{\Delta E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}(h_1 + V_1^2 / 2) = \dot{W}_{\text{out}} + \dot{m}(h_2 + V_2^2 / 2) \quad (\text{since } \dot{Q} \cong \Delta p e \cong 0)$$

$$\dot{W}_{\text{out}} = -\dot{m} \left(h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} \right)$$

Then the power output of the turbine is determined by substitution to be

$$\dot{W}_{\text{out}} = -(12 \text{ kg/s})(2392.5 - 3242.4 - 1.95) \text{ kJ/kg} = \mathbf{10.2 \text{ MW}}$$

(c) The inlet area of the turbine is determined from the mass flow rate relation,

$$\dot{m} = \frac{1}{v_1} A_1 V_1 \longrightarrow A_1 = \frac{\dot{m} v_1}{V_1} = \frac{(12 \text{ kg/s})(0.029782 \text{ m}^3/\text{kg})}{80 \text{ m/s}} = \mathbf{0.00447 \text{ m}^2}$$

5-51 Steam enters an adiabatic turbine at 10 MPa and 500°C and leaves at 10 kPa with a quality of 90 percent. Neglecting the changes in kinetic and potential energies, determine the mass flow rate required for a power output of 5 MW. *Answer: 4.852 kg/s*

Properties From the steam tables (Tables A-4 through 6)

$$\left. \begin{array}{l} P_1 = 10 \text{ MPa} \\ T_1 = 500^\circ\text{C} \end{array} \right\} h_1 = 3375.1 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_2 = 10 \text{ kPa} \\ x_2 = 0.90 \end{array} \right\} h_2 = h_f + x_2 h_{fg} = 191.81 + 0.90 \times 2392.1 = 2344.7 \text{ kJ/kg}$$

Analysis There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. We take the turbine as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\dot{\Delta E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

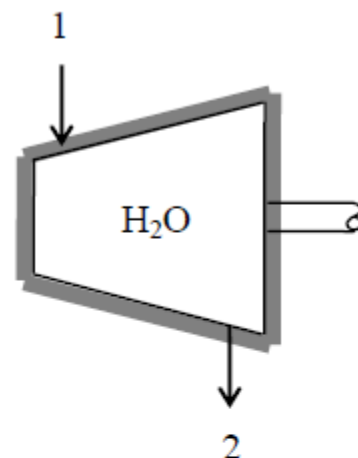
$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}h_1 = \dot{W}_{\text{out}} + \dot{m}h_2 \quad (\text{since } \dot{Q} \cong \Delta ke \cong \Delta pe \cong 0)$$

$$\dot{W}_{\text{out}} = -\dot{m}(h_2 - h_1)$$

Substituting, the required mass flow rate of the steam is determined to be

$$5000 \text{ kJ/s} = -\dot{m}(2344.7 - 3375.1) \text{ kJ/kg} \longrightarrow \dot{m} = \mathbf{4.852 \text{ kg/s}}$$



5-53 Steam enters an adiabatic turbine at 8 MPa and 500°C at a rate of 3 kg/s and leaves at 20 kPa. If the power output of the turbine is 2.5 MW, determine the temperature of the steam at the turbine exit. Neglect kinetic energy changes.

Answer: 60.1°C

5-54 Argon gas enters an adiabatic turbine steadily at 900 kPa and 450°C with a velocity of 80 m/s and leaves at 150 kPa with a velocity of 150 m/s. The inlet area of the turbine is 60 cm². If the power output of the turbine is 250 kW, determine the exit temperature of the argon.

$$\nu_1 = \frac{RT_1}{P_1} = \frac{(0.2081 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(723 \text{ K})}{900 \text{ kPa}} = 0.167 \text{ m}^3/\text{kg}$$

Thus,

$$\dot{m} = \frac{1}{\nu_1} A_1 V_1 = \frac{1}{0.167 \text{ m}^3/\text{kg}} (0.006 \text{ m}^2)(80 \text{ m/s}) = 2.874 \text{ kg/s}$$

We take the turbine as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\dot{\Delta E}_{system}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{m}(h_1 + V_1^2/2) = \dot{W}_{out} + \dot{m}(h_2 + V_2^2/2) \quad (\text{since } \dot{Q} \equiv \Delta pe \equiv 0)$$

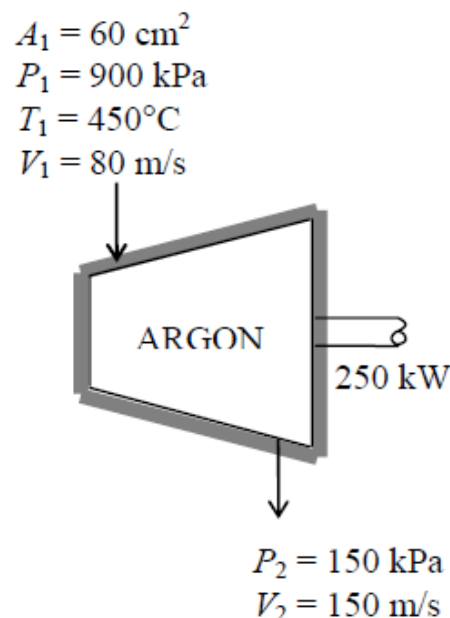
$$\dot{W}_{out} = -\dot{m} \left(h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} \right)$$

Substituting,

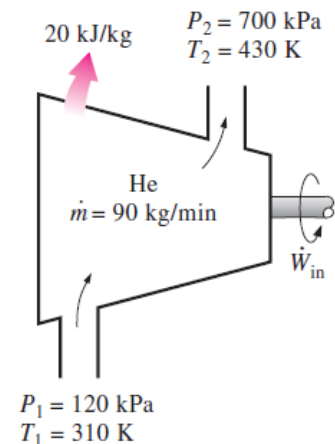
$$250 \text{ kJ/s} = -(2.874 \text{ kg/s}) \left[(0.5203 \text{ kJ/kg} \cdot ^\circ\text{C})(T_2 - 450^\circ\text{C}) + \frac{(150 \text{ m/s})^2 - (80 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) \right]$$

It yields

$$T_2 = 267.3^\circ\text{C}$$



5–60 Helium is to be compressed from 120 kPa and 310 K to 700 kPa and 430 K. A heat loss of 20 kJ/kg occurs during the compression process. Neglecting kinetic energy changes, determine the power input required for a mass flow rate of 90 kg/min.



5–61 Carbon dioxide enters an adiabatic compressor at 100 kPa and 300 K at a rate of 0.5 kg/s and leaves at 600 kPa and 450 K. Neglecting kinetic energy changes, determine (a) the volume flow rate of the carbon dioxide at the compressor inlet and (b) the power input to the compressor.

Answers: (a) $0.28 \text{ m}^3/\text{s}$, (b) 68.8 kW

Throttling Valves

5-67



Refrigerant-134a at 800 kPa and 25°C is throttled to a temperature of -20°C. Determine the pressure and the internal energy of the refrigerant at the final state. **Answers: 133 kPa, 80.7 kJ/kg**

Properties The inlet enthalpy of R-134a is, from the refrigerant tables (Tables A-11 through 13),

$$\left. \begin{array}{l} P_1 = 0.8 \text{ MPa} \\ T_1 = 25^\circ\text{C} \end{array} \right\} h_1 \cong h_{f@25^\circ\text{C}} = 86.41 \text{ kJ/kg}$$

Analysis There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. We take the throttling valve as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}h_1 = \dot{m}h_2$$

$$h_1 = h_2$$

since $\dot{Q} \cong \dot{W} = \Delta ke \cong \Delta pe \cong 0$. Then,

$$\left. \begin{array}{l} T_2 = -20^\circ\text{C} \\ (h_2 = h_1) \end{array} \right\} \begin{array}{ll} h_f = 25.49 \text{ kJ/kg}, & u_f = 25.39 \text{ kJ/kg} \\ h_g = 238.41 \text{ kJ/kg}, & u_g = 218.84 \text{ kJ/kg} \end{array}$$

Obviously $h_f < h_2 < h_g$, thus the refrigerant exists as a saturated mixture at the exit state, and thus

$$P_2 = P_{\text{sat}@-20^\circ\text{C}} = \mathbf{132.82 \text{ kPa}}$$

Also,

$$x_2 = \frac{h_2 - h_f}{h_{fg}} = \frac{86.41 - 25.49}{212.91} = 0.2861$$

Thus,

$$u_2 = u_f + x_2 u_{fg} = 25.39 + 0.2861 \times 193.45 = \mathbf{80.74 \text{ kJ/kg}}$$

$$P_1 = 0.8 \text{ MPa}$$

$$T_1 = 25^\circ\text{C}$$



R-134a

$$T_2 = -20^\circ\text{C}$$

5-68 A well-insulated valve is used to throttle steam from 8 MPa and 500°C to 6 MPa. Determine the final temperature of the steam. *Answer: 490.1°C*

Properties The inlet enthalpy of steam is (Tables A-6),

$$\left. \begin{array}{l} P_1 = 8 \text{ MPa} \\ T_1 = 500^\circ\text{C} \end{array} \right\} h_1 = 3399.5 \text{ kJ/kg}$$

Analysis There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. We take the throttling valve as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

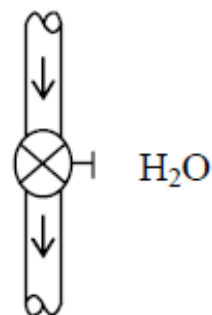
$$\dot{m}h_1 = \dot{m}h_2$$

$$h_1 = h_2$$

since $\dot{Q} \cong \dot{W} = \Delta ke \cong \Delta pe \cong 0$. Then the exit temperature of steam becomes

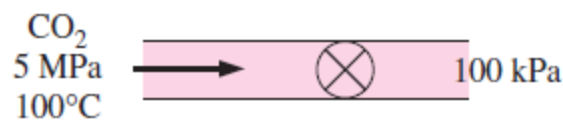
$$\left. \begin{array}{l} P_2 = 6 \text{ MPa} \\ (h_2 = h_1) \end{array} \right\} T_2 = \mathbf{490.1^\circ\text{C}}$$

$$\begin{array}{l} P_1 = 8 \text{ MPa} \\ T_1 = 500^\circ\text{C} \end{array}$$



$$P_2 = 6 \text{ MPa}$$

5-71 Carbon dioxide gas enters a throttling valve at 5 MPa and 100°C and leaves at 100 kPa. Determine the temperature change during this process if CO₂ is assumed to be (a) an ideal gas and (b) a real gas.



5-75 A hot-water stream at 80°C enters a mixing chamber with a mass flow rate of 0.5 kg/s where it is mixed with a stream of cold water at 20°C . If it is desired that the mixture leave the chamber at 42°C , determine the mass flow rate of the cold-water stream. Assume all the streams are at a pressure of 250 kPa . **Answer:** 0.865 kg/s

Properties Noting that $T < T_{\text{sat @ } 250\text{ kPa}} = 127.41^\circ\text{C}$, the water in all three streams exists as a compressed liquid, which can be approximated as a saturated liquid at the given temperature. Thus,

$$h_1 \cong h_f @ 80^\circ\text{C} = 335.02\text{ kJ/kg}$$

$$h_2 \cong h_f @ 20^\circ\text{C} = 83.915\text{ kJ/kg}$$

$$h_3 \cong h_f @ 42^\circ\text{C} = 175.90\text{ kJ/kg}$$

Analysis We take the mixing chamber as the system, which is a control volume. The mass and energy balances for this steady-flow system can be expressed in the rate form as

Mass balance:

$$\dot{m}_{\text{in}} - \dot{m}_{\text{out}} = \Delta \dot{m}_{\text{system}} \overset{\text{no (steady)}}{=} 0 \longrightarrow \dot{m}_1 + \dot{m}_2 = \dot{m}_3$$

Energy balance:

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \overset{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3 \quad (\text{since } \dot{Q} = \dot{W} = \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

Combining the two relations and solving for $\dot{m}_2$ gives

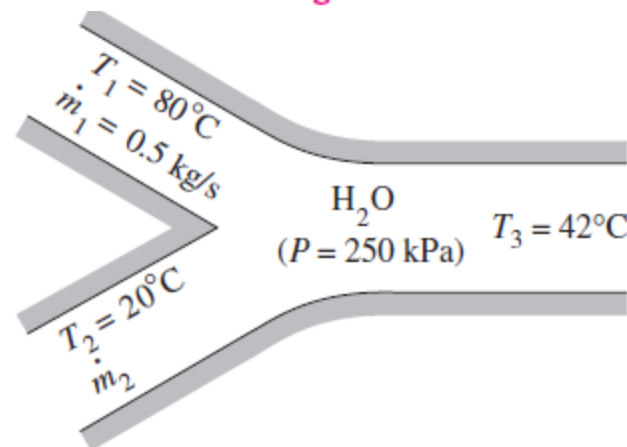
$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = (\dot{m}_1 + \dot{m}_2) h_3$$

$$\dot{m}_2 = \frac{h_1 - h_3}{h_3 - h_2} \dot{m}_1$$

Substituting, the mass flow rate of cold water stream is determined to be

$$\dot{m}_2 = \frac{(335.02 - 175.90)\text{ kJ/kg}}{(175.90 - 83.915)\text{ kJ/kg}} (0.5\text{ kg/s}) = \mathbf{0.865\text{ kg/s}}$$

Mixing Chambers and Heat Exchangers



5-79 A stream of refrigerant-134a at 1 MPa and 12°C is mixed with another stream at 1 MPa and 60°C. If the mass flow rate of the cold stream is twice that of the hot one, determine the temperature and the quality of the exit stream.

Mass balance:

$$\dot{m}_{\text{in}} - \dot{m}_{\text{out}} = \Delta \dot{m}_{\text{system}} \overset{\text{no}}{\neq} 0 \text{ (steady)} = 0$$

$$\dot{m}_{\text{in}} = \dot{m}_{\text{out}}$$

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3 = 3\dot{m}_2 \text{ since } \dot{m}_1 = 2\dot{m}_2$$

Energy balance:

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \overset{\text{no}}{\neq} 0 \text{ (steady)} = 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3 \text{ (since } \dot{Q} = \dot{W} = \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

Combining the two gives $2\dot{m}_2 h_1 + \dot{m}_2 h_2 = 3\dot{m}_2 h_3$ or $h_3 = (2h_1 + h_2)/3$

Substituting,

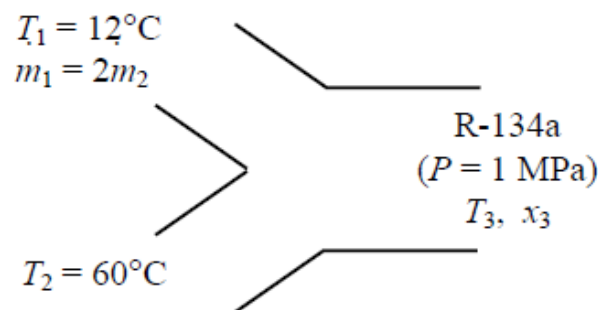
$$h_3 = (2 \times 68.18 + 293.38)/3 = 143.25 \text{ kJ/kg}$$

At 1 MPa, $h_f = 107.32 \text{ kJ/kg}$ and $h_g = 270.99 \text{ kJ/kg}$. Thus the exit stream is a saturated mixture since $h_f < h_3 < h_g$. Therefore,

$$T_3 = T_{\text{sat @ 1 MPa}} = 39.37^\circ\text{C}$$

and

$$x_3 = \frac{h_3 - h_f}{h_{fg}} = \frac{143.25 - 107.32}{163.67} = 0.220$$



5-83 Refrigerant-134a at 700 kPa, 70°C, and 8 kg/min is cooled by water in a condenser until it exists as a saturated liquid at the same pressure. The cooling water enters the condenser at 300 kPa and 15°C and leaves at 25°C at the same pressure. Determine the mass flow rate of the cooling water required to cool the refrigerant. *Answer: 42.0 kg/min*

Properties The enthalpies of R-134a at the inlet and the exit states are (Tables A-11 through A-13)

$$\left. \begin{array}{l} P_3 = 700 \text{ kPa} \\ T_3 = 70^\circ\text{C} \end{array} \right\} h_3 = 308.33 \text{ kJ/kg}$$

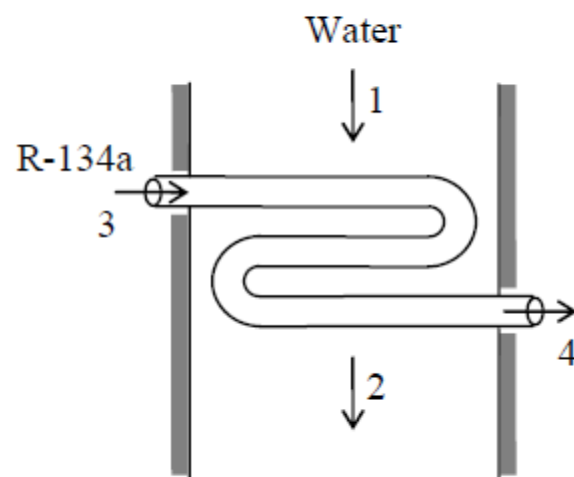
$$\left. \begin{array}{l} P_4 = 700 \text{ kPa} \\ \text{sat. liquid} \end{array} \right\} h_4 = h_f @ 700 \text{ kPa} = 88.82 \text{ kJ/kg}$$

Water exists as compressed liquid at both states, and thus (Table A-4)

$$h_1 \cong h_f @ 15^\circ\text{C} = 62.98 \text{ kJ/kg}$$

$$h_2 \cong h_f @ 25^\circ\text{C} = 104.83 \text{ kJ/kg}$$

Analysis We take the heat exchanger as the system, which is a control volume. The mass and energy balances for this steady-flow system can be expressed in the rate form as



Mass balance (for each fluid stream):

$$\dot{m}_{\text{in}} - \dot{m}_{\text{out}} = \Delta \dot{m}_{\text{system}} \overset{\text{no (steady)}}{=} 0 \rightarrow \dot{m}_{\text{in}} = \dot{m}_{\text{out}} \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}_w \text{ and } \dot{m}_3 = \dot{m}_4 = \dot{m}_R$$

Energy balance (for the heat exchanger):

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \overset{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}_1 h_1 + \dot{m}_3 h_3 = \dot{m}_2 h_2 + \dot{m}_4 h_4 \quad (\text{since } \dot{Q} = \dot{W} = \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

Combining the two, $\dot{m}_w (h_2 - h_1) = \dot{m}_R (h_3 - h_4)$

Solving for $\dot{m}_w$:

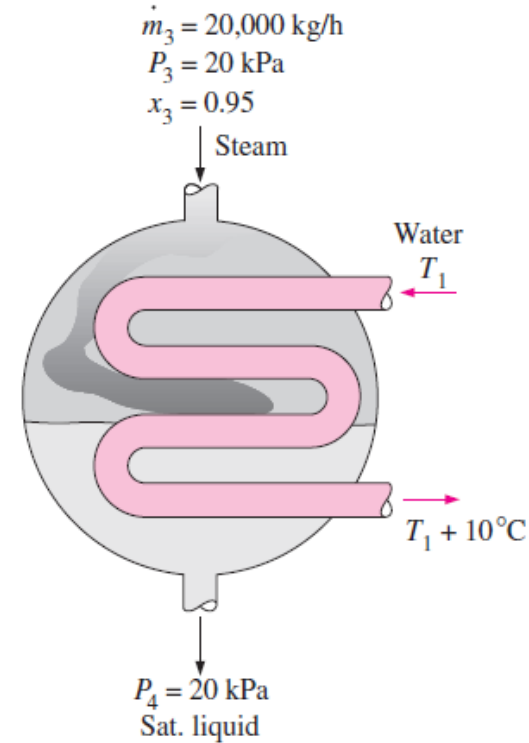
$$\dot{m}_w = \frac{h_3 - h_4}{h_2 - h_1} \dot{m}_R$$

Substituting,

$$\dot{m}_w = \frac{(308.33 - 88.82) \text{ kJ/kg}}{(104.83 - 62.98) \text{ kJ/kg}} (8 \text{ kg/min}) = \mathbf{42.0 \text{ kg/min}}$$

5-85 Steam enters the condenser of a steam power plant at 20 kPa and a quality of 95 percent with a mass flow rate of 20,000 kg/h. It is to be cooled by water from a nearby river by circulating the water through the tubes within the condenser. To prevent thermal pollution, the river water is not allowed to experience a temperature rise above 10°C. If the steam is to leave the condenser as saturated liquid at 20 kPa, determine the mass flow rate of the cooling water required.

Answer: 297.7 kg/s



5-91 A thin-walled double-pipe counter-flow heat exchanger is used to cool oil ($c_p = 2.20 \text{ kJ/kg} \cdot ^\circ\text{C}$) from 150°C to 40°C at a rate of 2 kg/s by water ($c_p = 4.18 \text{ kJ/kg} \cdot ^\circ\text{C}$) that enters at 22°C at a rate of 1.5 kg/s . Determine the rate of heat transfer in the heat exchanger and the exit temperature of water.

Properties The specific heats of water and oil are given to be 4.18 and $2.20 \text{ kJ/kg} \cdot ^\circ\text{C}$, respectively.

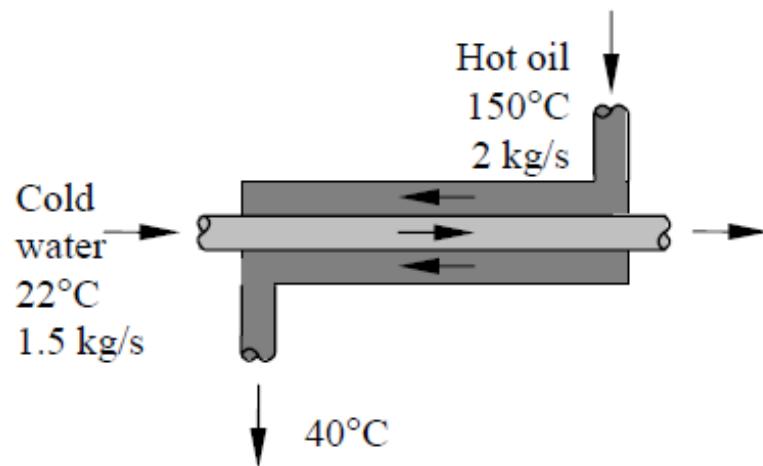
Analysis We take the oil tubes as the system, which is a control volume. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\approx 0 \text{ (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}h_1 = \dot{Q}_{\text{out}} + \dot{m}h_2 \quad (\text{since } \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

$$\dot{Q}_{\text{out}} = \dot{m}c_p(T_1 - T_2)$$



Then the rate of heat transfer from the oil becomes

$$\dot{Q} = [\dot{m}c_p(T_{\text{in}} - T_{\text{out}})]_{\text{oil}} = (2 \text{ kg/s})(2.2 \text{ kJ/kg} \cdot ^\circ\text{C})(150^\circ\text{C} - 40^\circ\text{C}) = \mathbf{484 \text{ kW}}$$

Noting that the heat lost by the oil is gained by the water, the outlet temperature of the water is determined from

$$\dot{Q} = [\dot{m}c_p(T_{\text{out}} - T_{\text{in}})]_{\text{water}} \longrightarrow T_{\text{out}} = T_{\text{in}} + \frac{\dot{Q}}{\dot{m}_{\text{water}}c_p} = 22^\circ\text{C} + \frac{484 \text{ kJ/s}}{(1.5 \text{ kg/s})(4.18 \text{ kJ/kg} \cdot ^\circ\text{C})} = \mathbf{99.2^\circ\text{C}}$$

5-92 Cold water ($c_p = 4.18 \text{ kJ/kg} \cdot ^\circ\text{C}$) leading to a shower enters a thin-walled double-pipe counter-flow heat exchanger at 15°C at a rate of 0.60 kg/s and is heated to 45°C by hot water ($c_p = 4.19 \text{ kJ/kg} \cdot ^\circ\text{C}$) that enters at 100°C at a rate of 3 kg/s . Determine the rate of heat transfer in the heat exchanger and the exit temperature of the hot water.

5-93 Air ($c_p = 1.005 \text{ kJ/kg} \cdot ^\circ\text{C}$) is to be preheated by hot exhaust gases in a cross-flow heat exchanger before it enters the furnace. Air enters the heat exchanger at 95 kPa and 20°C at a rate of $0.8 \text{ m}^3/\text{s}$. The combustion gases ($c_p = 1.10 \text{ kJ/kg} \cdot ^\circ\text{C}$) enter at 180°C at a rate of 1.1 kg/s and leave at 95°C . Determine the rate of heat transfer to the air and its outlet temperature.

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0 \rightarrow \dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}h_1 = \dot{Q}_{\text{out}} + \dot{m}h_2 \quad (\text{since } \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

$$\dot{Q}_{\text{out}} = \dot{m}c_p(T_1 - T_2)$$

Then the rate of heat transfer from the exhaust gases becomes

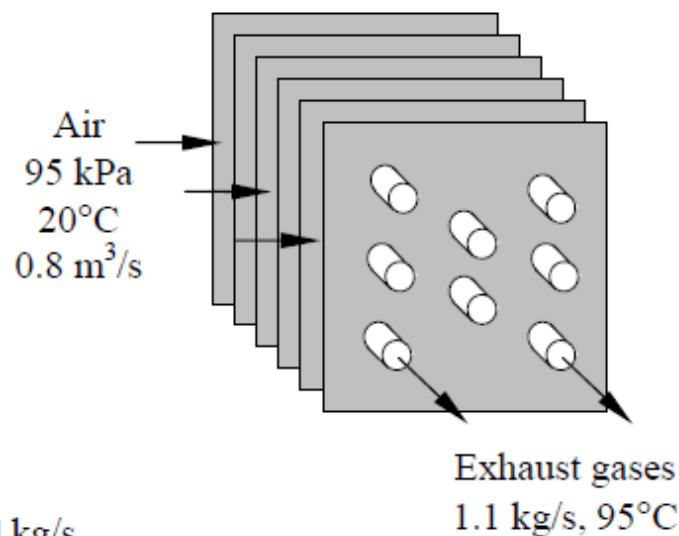
$$\begin{aligned} \dot{Q} &= [\dot{m}c_p(T_{\text{in}} - T_{\text{out}})]_{\text{gas}} \\ &= (1.1 \text{ kg/s})(1.1 \text{ kJ/kg} \cdot ^\circ\text{C})(180^\circ\text{C} - 95^\circ\text{C}) \\ &= \mathbf{102.85 \text{ kW}} \end{aligned}$$

The mass flow rate of air is

$$\dot{m} = \frac{P\dot{V}}{RT} = \frac{(95 \text{ kPa})(0.8 \text{ m}^3/\text{s})}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}) \times 293 \text{ K}} = 0.904 \text{ kg/s}$$

Noting that heat loss by the exhaust gases is equal to the heat gain by the air, the outlet temperature of the air becomes

$$\dot{Q} = \dot{m}c_p(T_{\text{c,out}} - T_{\text{c,in}}) \longrightarrow T_{\text{c,out}} = T_{\text{c,in}} + \frac{\dot{Q}}{\dot{m}c_p} = 20^\circ\text{C} + \frac{102.85 \text{ kW}}{(0.904 \text{ kg/s})(1.005 \text{ kJ/kg} \cdot ^\circ\text{C})} = \mathbf{133.2^\circ\text{C}}$$



5-94 A well-insulated shell-and-tube heat exchanger is used to heat water ($c_p = 4.18 \text{ kJ/kg} \cdot ^\circ\text{C}$) in the tubes from 20 to 70°C at a rate of 4.5 kg/s . Heat is supplied by hot oil ($c_p = 2.30 \text{ kJ/kg} \cdot ^\circ\text{C}$) that enters the shell side at 170°C at a rate of 10 kg/s . Determine the rate of heat transfer in the heat exchanger and the exit temperature of oil.

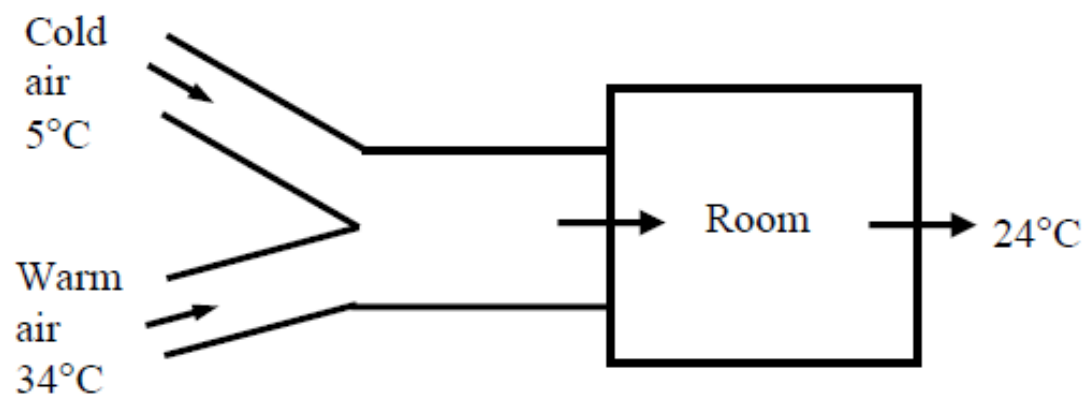
5–96 An air-conditioning system involves the mixing of cold air and warm outdoor air before the mixture is routed to the conditioned room in steady operation. Cold air enters the mixing chamber at 5°C and 105 kPa at a rate of $1.25\text{ m}^3/\text{s}$ while warm air enters at 34°C and 105 kPa . The air leaves the room at 24°C . The ratio of the mass flow rates of the hot to cold air streams is 1.6 . Using variable specific heats, determine (a) the mixture temperature at the inlet of the room and (b) the rate of heat gain of the room.

Properties The gas constant of air is $R = 0.287\text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$. The enthalpies of air are obtained from air table (Table A-17) as

$$h_1 = h_{@278\text{ K}} = 278.13\text{ kJ/kg}$$

$$h_2 = h_{@307\text{ K}} = 307.23\text{ kJ/kg}$$

$$h_{\text{room}} = h_{@297\text{ K}} = 297.18\text{ kJ/kg}$$



Analysis (a) We take the mixing chamber as the system, which is a control volume since mass crosses the boundary. The mass and energy balances for this steady-flow system can be expressed in the rate form as

Mass balance:

$$\dot{m}_{\text{in}} - \dot{m}_{\text{out}} = \Delta \dot{m}_{\text{system}} \xrightarrow{\text{steady}} 0 \rightarrow \dot{m}_{\text{in}} = \dot{m}_{\text{out}} \rightarrow \dot{m}_1 + 1.6\dot{m}_1 = \dot{m}_3 = 2.6\dot{m}_1 \quad \text{since} \quad \dot{m}_2 = 1.6\dot{m}_1$$



Energy balance:

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \overset{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3 \quad (\text{since } \dot{Q} \cong \dot{W} \cong \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

Combining the two gives $\dot{m}_1 h_1 + 1.6 \dot{m}_1 h_2 = 2.6 \dot{m}_1 h_3$ or $h_3 = (h_1 + 1.6 h_2) / 2.6$

Substituting,

$$h_3 = (278.13 + 1.6 \times 307.23) / 2.6 = 296.04 \text{ kJ/kg}$$

From air table at this enthalpy, the mixture temperature is

$$T_3 = T_{@h=296.04 \text{ kJ/kg}} = 295.9 \text{ K} = \mathbf{22.9^\circ\text{C}}$$

(b) The mass flow rates are determined as follows

$$v_1 = \frac{RT_1}{P} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(5 + 273 \text{ K})}{105 \text{ kPa}} = 0.7599 \text{ m}^3/\text{kg}$$

$$\dot{m}_1 = \frac{\dot{V}_1}{v_1} = \frac{1.25 \text{ m}^3/\text{s}}{0.7599 \text{ m}^3/\text{kg}} = 1.645 \text{ kg/s}$$

$$\dot{m}_3 = 2.6 \dot{m}_1 = 2.6(1.645 \text{ kg/s}) = 4.277 \text{ kg/s}$$

The rate of heat gain of the room is determined from

$$\dot{Q}_{\text{cool}} = \dot{m}_3 (h_{\text{room}} - h_3) = (4.277 \text{ kg/s})(297.18 - 296.04) \text{ kJ/kg} = \mathbf{4.88 \text{ kW}}$$

5-97 Hot exhaust gases of an internal combustion engine are to be used to produce saturated water vapor at 2 MPa pressure. The exhaust gases enter the heat exchanger at 400°C at a rate of 32 kg/min while water enters at 15°C. The heat exchanger is not well insulated, and it is estimated that 10 percent of heat given up by the exhaust gases is lost to the surroundings. If the mass flow rate of the exhaust gases is 15 times that of the water, determine (a) the temperature of the exhaust gases at the heat exchanger exit and (b) the rate of heat transfer to the water. Use the constant specific heat properties of air for the exhaust gases.

5-98 A desktop computer is to be cooled by a fan. The electronic components of the computer consume 60 W of power under full-load conditions. The computer is to operate in environments at temperatures up to 45°C and at elevations up to 3400 m where the average atmospheric pressure is 66.63 kPa. The exit temperature of air is not to exceed 60°C to meet the reliability requirements. Also, the average velocity of air is not to exceed 110 m/min at the exit of the computer case where the fan is installed to keep the noise level down. Determine the flow rate of the fan that needs to be installed and the diameter of the casing of the fan.

Properties The specific heat of air at the average temperature of $T_{\text{avg}} = (45+60)/2 = 52.5^\circ\text{C} = 325.5\text{ K}$ is $c_p = 1.0065\text{ kJ/kg}\cdot^\circ\text{C}$. The gas constant for air is $R = 0.287\text{ kJ/kg}\cdot\text{K}$ (Table A-2).

Analysis The fan selected must be able to meet the cooling requirements of the computer at worst conditions. Therefore, we assume air to enter the computer at 66.63 kPa and 45°C, and leave at 60°C.

We take the air space in the computer as the system, which is a control volume. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \overset{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{Q}_{\text{in}} + \dot{m}h_1 = \dot{m}h_2 \quad (\text{since } \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

$$\dot{Q}_{\text{in}} = \dot{m}c_p(T_2 - T_1)$$

Then the required mass flow rate of air to absorb heat at a rate of 60 W is determined to be



$$\dot{Q} = \dot{m}c_p(T_{\text{out}} - T_{\text{in}}) \rightarrow \dot{m} = \frac{\dot{Q}}{c_p(T_{\text{out}} - T_{\text{in}})} = \frac{60 \text{ W}}{(1006.5 \text{ J/kg} \cdot ^\circ\text{C})(60 - 45)^\circ\text{C}} \\ = 0.00397 \text{ kg/s} = 0.238 \text{ kg/min}$$

The density of air entering the fan at the exit and its volume flow rate are

$$\rho = \frac{P}{RT} = \frac{66.63 \text{ kPa}}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(60 + 273)\text{K}} = 0.6972 \text{ kg/m}^3$$

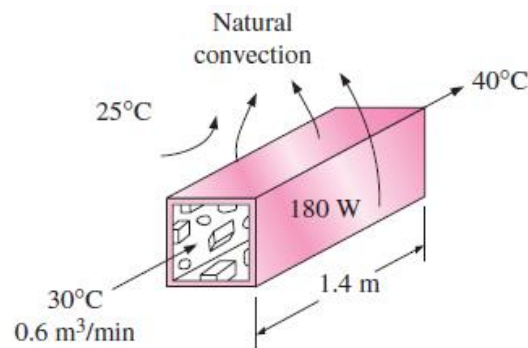
$$\dot{V} = \frac{\dot{m}}{\rho} = \frac{0.238 \text{ kg/min}}{0.6972 \text{ kg/m}^3} = \mathbf{0.341 \text{ m}^3/\text{min}}$$

For an average exit velocity of 110 m/min, the diameter of the casing of the fan is determined from

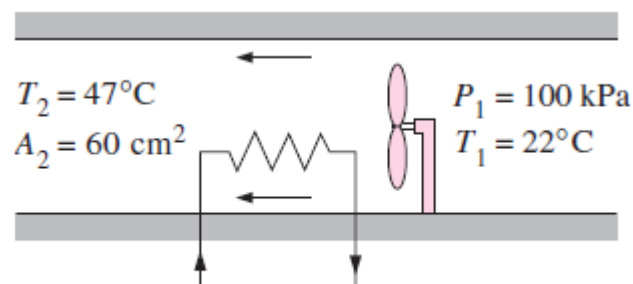
$$\dot{V} = A_c V = \frac{\pi D^2}{4} V \rightarrow D = \sqrt{\frac{4\dot{V}}{\pi V}} = \sqrt{\frac{(4)(0.341 \text{ m}^3/\text{min})}{\pi(110 \text{ m/min})}} = 0.063 \text{ m} = \mathbf{6.3 \text{ cm}}$$

5-99 Repeat Prob. 5-98 for a computer that consumes 100 W of power.

5-105  The components of an electronic system dissipating 180 W are located in a 1.4-m-long horizontal duct whose cross section is 20 cm $\times$ 20 cm. The components in the duct are cooled by forced air that enters the duct at 30°C and 1 atm at a rate of 0.6 m³/min and leaves at 40°C. Determine the rate of heat transfer from the outer surfaces of the duct to the ambient. **Answer: 63 W**



5-114 A hair dryer is basically a duct in which a few layers of electric resistors are placed. A small fan pulls the air in and forces it through the resistors where it is heated. Air enters a 1200-W hair dryer at 100 kPa and 22°C and leaves at 47°C. The cross-sectional area of the hair dryer at the exit is 60 cm². Neglecting the power consumed by the fan and the heat losses through the walls of the hair dryer, determine (a) the volume flow rate of air at the inlet and (b) the velocity of the air at the exit. **Answers: (a) 0.0404 m³/s, (b) 7.31 m/s**



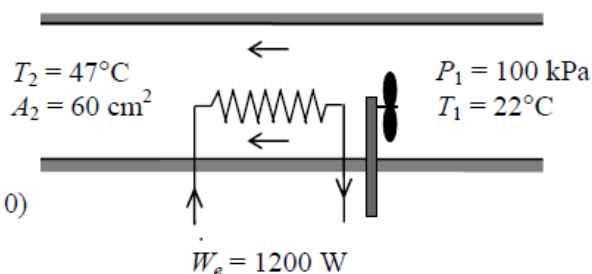
Analysis We take the *hair dryer* as the system, which is a control volume since mass crosses the boundary. There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{W}_{\text{e,in}} + \dot{m}h_1 = \dot{m}h_2 \quad (\text{since } \dot{Q}_{\text{out}} \cong \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

$$\dot{W}_{\text{e,in}} = \dot{m}(h_2 - h_1) = \dot{m}c_p(T_2 - T_1)$$



Substituting, the mass and volume flow rates of air are determined to be

$$\dot{m} = \frac{\dot{W}_{\text{e,in}}}{c_p(T_2 - T_1)} = \frac{1.2 \text{ kJ/s}}{(1.005 \text{ kJ/kg} \cdot ^\circ\text{C})(47 - 22)^\circ\text{C}} = 0.04776 \text{ kg/s}$$

$$\nu_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(295 \text{ K})}{(100 \text{ kPa})} = 0.8467 \text{ m}^3/\text{kg}$$

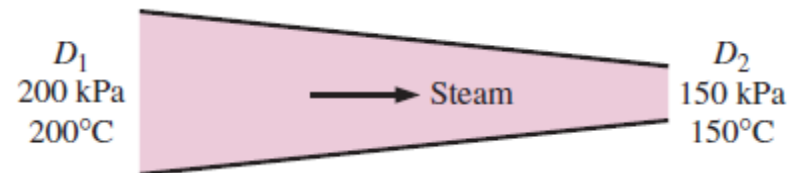
$$\dot{V}_1 = \dot{m}\nu_1 = (0.04776 \text{ kg/s})(0.8467 \text{ m}^3/\text{kg}) = \mathbf{0.0404 \text{ m}^3/\text{s}}$$

(b) The exit velocity of air is determined from the mass balance $\dot{m}_1 = \dot{m}_2 = \dot{m}$ to be

$$\nu_2 = \frac{RT_2}{P_2} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(320 \text{ K})}{(100 \text{ kPa})} = 0.9184 \text{ m}^3/\text{kg}$$

$$\dot{m} = \frac{1}{\nu_2} A_2 V_2 \longrightarrow V_2 = \frac{\dot{m}\nu_2}{A_2} = \frac{(0.04776 \text{ kg/s})(0.9184 \text{ m}^3/\text{kg})}{60 \times 10^{-4} \text{ m}^2} = \mathbf{7.31 \text{ m/s}}$$

5-120 Steam enters an insulated pipe at 200 kPa and 200°C and leaves at 150 kPa and 150°C. The inlet-to-outlet diameter ratio for the pipe is $D_1/D_2 = 1.80$. Determine the inlet and exit velocities of the steam.



5-127 A 2-m³ rigid tank initially contains air at 100 kPa and 22°C. The tank is connected to a supply line through a valve. Air is flowing in the supply line at 600 kPa and 22°C. The valve is opened, and air is allowed to enter the tank until the pressure in the tank reaches the line pressure, at which point the valve is closed. A thermometer placed in the tank indicates that the air temperature at the final state is 77°C. Determine (a) the mass of air that has entered the tank and (b) the amount of heat transfer. *Answers: (a) 9.58 kg, (b) $Q_{\text{out}} = 339 \text{ kJ}$*

Analysis (a) We take the tank as the system, which is a control volume since mass crosses the boundary. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances for this uniform-flow system can be expressed as

Mass balance:

$$m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_i = m_2 - m_1$$

Energy balance:

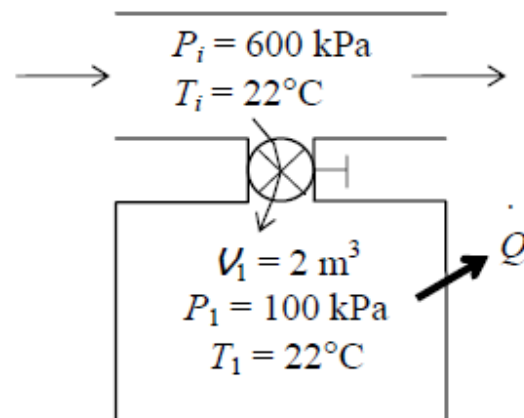
$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} + m_i h_i = m_2 u_2 - m_1 u_1 \quad (\text{since } W \cong ke \cong pe \cong 0)$$

The initial and the final masses in the tank are

$$m_1 = \frac{P_1 V}{RT_1} = \frac{(100 \text{ kPa})(2 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(295 \text{ K})} = 2.362 \text{ kg}$$

$$m_2 = \frac{P_2 V}{RT_2} = \frac{(600 \text{ kPa})(2 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(350 \text{ K})} = 11.946 \text{ kg}$$



Then from the mass balance,

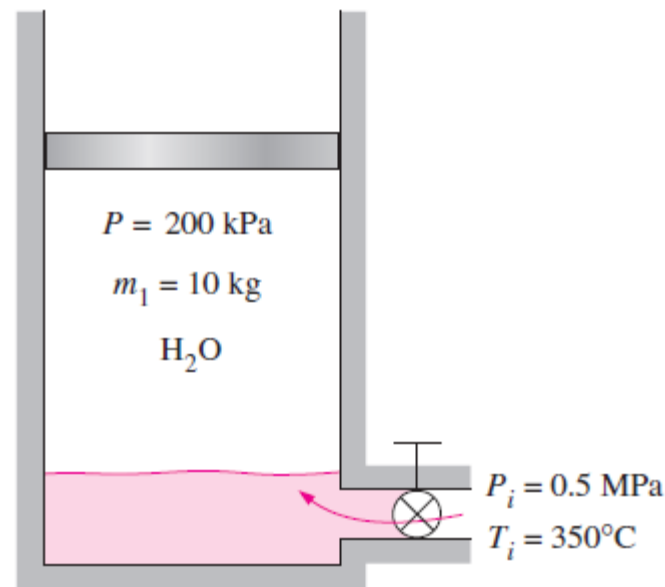
$$m_i = m_2 - m_1 = 11.946 - 2.362 = \mathbf{9.584 \text{ kg}}$$

(b) The heat transfer during this process is determined from

$$\begin{aligned} Q_{\text{in}} &= -m_i h_i + m_2 u_2 - m_1 u_1 \\ &= -(9.584 \text{ kg})(295.17 \text{ kJ/kg}) + (11.946 \text{ kg})(250.02 \text{ kJ/kg}) - (2.362 \text{ kg})(210.49 \text{ kJ/kg}) \\ &= -339 \text{ kJ} \rightarrow Q_{\text{out}} = \mathbf{339 \text{ kJ}} \end{aligned}$$

Discussion The negative sign for heat transfer indicates that the assumed direction is wrong. Therefore, we reversed the direction.

5-131 An insulated, vertical piston–cylinder device initially contains 10 kg of water, 6 kg of which is in the vapor phase. The mass of the piston is such that it maintains a constant pressure of 200 kPa inside the cylinder. Now steam at 0.5 MPa and 350°C is allowed to enter the cylinder from a supply line until all the liquid in the cylinder has vaporized. Determine (a) the final temperature in the cylinder and (b) the mass of the steam that has entered. *Answers: (a) 120.2°C, (b) 19.07 kg*

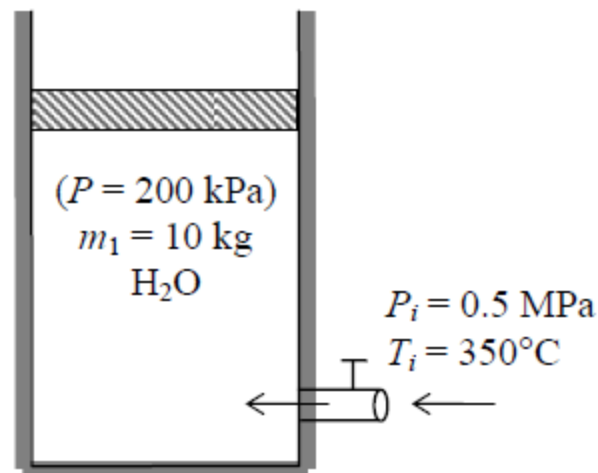


Properties The properties of steam are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa} \\ x_1 = 0.6 \end{array} \right\} h_1 = h_f + x_1 h_{fg} = 504.71 + 0.6 \times 2201.6 = 1825.6 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_2 = 200 \text{ kPa} \\ \text{sat. vapor} \end{array} \right\} h_2 = h_{g@200 \text{ kPa}} = 2706.3 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_i = 0.5 \text{ MPa} \\ T_i = 350^\circ\text{C} \end{array} \right\} h_i = 3168.1 \text{ kJ/kg}$$



Analysis (a) The cylinder contains saturated vapor at the final state at a pressure of 200 kPa, thus the final temperature in the cylinder must be

$$T_2 = T_{\text{sat}@200 \text{ kPa}} = 120.2^\circ\text{C}$$

(b) We take the cylinder as the system, which is a control volume since mass crosses the boundary. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances for this uniform-flow system can be expressed as

Mass balance: $m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_i = m_2 - m_1$

Energy balance:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc. energies}}}$$

$$m_i h_i = W_{\text{b,out}} + m_2 u_2 - m_1 u_1 \quad (\text{since } Q \cong ke \cong pe \cong 0)$$

Combining the two relations gives $0 = W_{\text{b,out}} - (m_2 - m_1)h_i + m_2 u_2 - m_1 u_1$

or,
$$0 = -(m_2 - m_1)h_i + m_2 h_2 - m_1 h_1$$

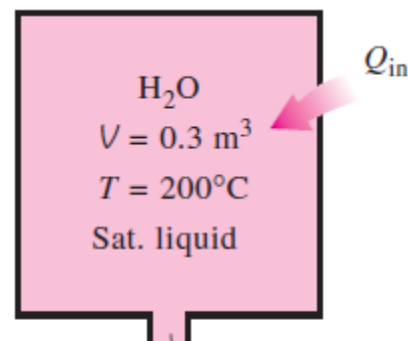
since the boundary work and ΔU combine into ΔH for constant pressure expansion and compression processes. Solving for m_2 and substituting,

$$m_2 = \frac{h_i - h_1}{h_i - h_2} m_1 = \frac{(3168.1 - 1825.6) \text{ kJ/kg}}{(3168.1 - 2706.3) \text{ kJ/kg}} (10 \text{ kg}) = 29.07 \text{ kg}$$

Thus,

$$m_i = m_2 - m_1 = 29.07 - 10 = \mathbf{19.07 \text{ kg}}$$

5-133 A 0.3-m³ rigid tank is filled with saturated liquid water at 200°C. A valve at the bottom of the tank is opened, and liquid is withdrawn from the tank. Heat is transferred to the water such that the temperature in the tank remains constant. Determine the amount of heat that must be transferred by the time one-half of the total mass has been withdrawn.



Properties The properties of water are (Tables A-4 through A-6)

$$\left. \begin{array}{l} T_1 = 200^\circ\text{C} \\ \text{sat. liquid} \end{array} \right\} \begin{array}{l} v_1 = v_{f@200^\circ\text{C}} = 0.001157 \text{ m}^3/\text{kg} \\ u_1 = u_{f@200^\circ\text{C}} = 850.46 \text{ kJ/kg} \end{array}$$

$$\left. \begin{array}{l} T_e = 200^\circ\text{C} \\ \text{sat. liquid} \end{array} \right\} h_e = h_{f@200^\circ\text{C}} = 852.26 \text{ kJ/kg}$$

Analysis We take the tank as the system, which is a control volume since mass crosses the boundary. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances for this uniform-flow system can be expressed as

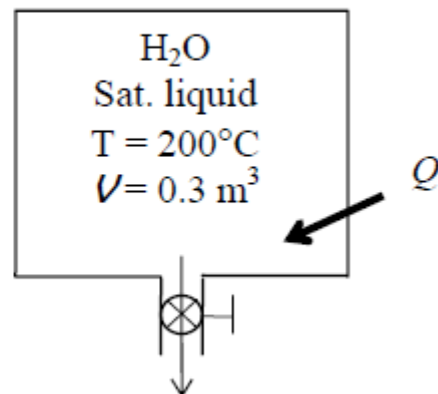
Mass balance: $m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_e = m_1 - m_2$

Energy balance:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} = m_e h_e + m_2 u_2 - m_1 u_1 \quad (\text{since } W \cong ke \cong pe \cong 0)$$

The initial and the final masses in the tank are



$$m_1 = \frac{V_1}{v_1} = \frac{0.3 \text{ m}^3}{0.001157 \text{ m}^3/\text{kg}} = 259.4 \text{ kg}$$

$$m_2 = \frac{1}{2} m_1 = \frac{1}{2} (259.4 \text{ kg}) = 129.7 \text{ kg}$$

Then from the mass balance,

$$m_e = m_1 - m_2 = 259.4 - 129.7 = 129.7 \text{ kg}$$

Now we determine the final internal energy,

$$v_2 = \frac{V}{m_2} = \frac{0.3 \text{ m}^3}{129.7 \text{ kg}} = 0.002313 \text{ m}^3/\text{kg}$$

$$x_2 = \frac{v_2 - v_f}{v_{fg}} = \frac{0.002313 - 0.001157}{0.12721 - 0.001157} = 0.009171$$

$$\left. \begin{array}{l} T_2 = 200^\circ\text{C} \\ x_2 = 0.009171 \end{array} \right\} u_2 = u_f + x_2 u_{fg} = 850.46 + (0.009171)(1743.7) = 866.46 \text{ kJ/kg}$$

Then the heat transfer during this process is determined from the energy balance by substitution to be

$$\begin{aligned} Q &= (129.7 \text{ kg})(852.26 \text{ kJ/kg}) + (129.7 \text{ kg})(866.46 \text{ kJ/kg}) - (259.4 \text{ kg})(850.46 \text{ kJ/kg}) \\ &= 2308 \text{ kJ} \end{aligned}$$

5-141 A balloon initially contains 65 m³ of helium gas at atmospheric conditions of 100 kPa and 22°C. The balloon is connected by a valve to a large reservoir that supplies helium gas at 150 kPa and 25°C. Now the valve is opened, and helium is allowed to enter the balloon until pressure equilibrium with the helium at the supply line is reached. The material of the balloon is such that its volume increases linearly with pressure. If no heat transfer takes place during this process, determine the final temperature in the balloon. *Answer: 334 K*

Analysis We take the cylinder as the system, which is a control volume since mass crosses the boundary. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances for this uniform-flow system can be expressed as

Mass balance:

$$m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_i = m_2 - m_1$$

Energy balance:

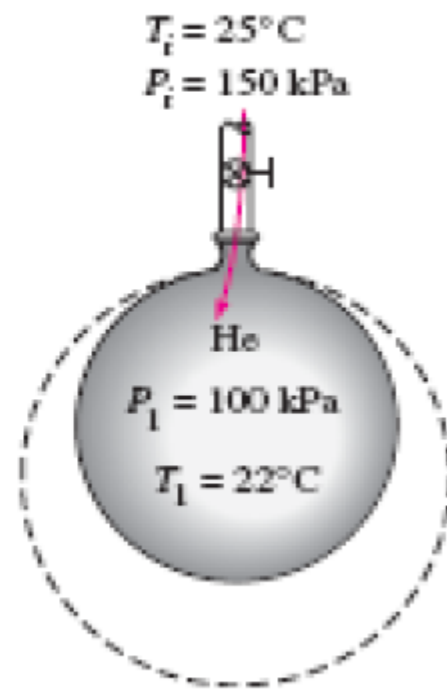
$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$m_i h_i = W_{b,\text{out}} + m_2 u_2 - m_1 u_1 \quad (\text{since } Q \cong ke \cong pe \cong 0)$$

$$m_1 = \frac{P_1 V_1}{RT_1} = \frac{(100 \text{ kPa})(65 \text{ m}^3)}{(2.0769 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(295 \text{ K})} = 10.61 \text{ kg}$$

$$\frac{P_1}{P_2} = \frac{V_1}{V_2} \longrightarrow V_2 = \frac{P_1}{P_2} V_1 = \frac{150 \text{ kPa}}{100 \text{ kPa}} (65 \text{ m}^3) = 97.5 \text{ m}^3$$

$$m_2 = \frac{P_2 V_2}{RT_2} = \frac{(150 \text{ kPa})(97.5 \text{ m}^3)}{(2.0769 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(T_2 \text{ K})} = \frac{7041.74}{T_2} \text{ kg}$$



Then from the mass balance,

$$m_i = m_2 - m_1 = \frac{7041.74}{T_2} - 10.61 \text{ kg}$$

Noting that P varies linearly with $\mathcal{V}$, the boundary work done during this process is

$$W_b = \frac{P_1 + P_2}{2} (\mathcal{V}_2 - \mathcal{V}_1) = \frac{(100 + 150) \text{ kPa}}{2} (97.5 - 65) \text{ m}^3 = 4062.5 \text{ kJ}$$

Using specific heats, the energy balance relation reduces to

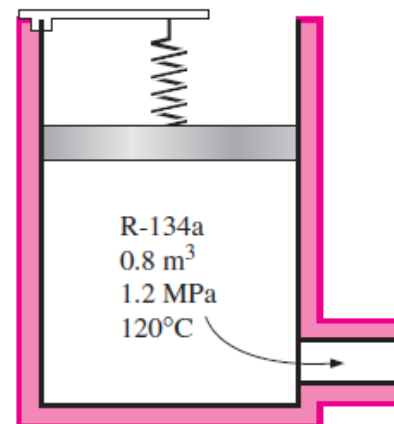
$$W_{b,\text{out}} = m_i c_p T_i - m_2 c_v T_2 + m_1 c_v T_1$$

Substituting,

$$4062.5 = \left(\frac{7041.74}{T_2} - 10.61 \right) (5.1926)(298) - \frac{7041.74}{T_2} (3.1156) T_2 + (10.61)(3.1156)(295)$$

It yields $T_2 = \mathbf{333.6 \text{ K}}$

5-142 An insulated vertical piston–cylinder device initially contains 0.8 m^3 of refrigerant-134a at 1.2 MPa and 120°C . A linear spring at this point applies full force to the piston. A valve connected to the cylinder is now opened, and refrigerant is allowed to escape. The spring unwinds as the piston moves down, and the pressure and volume drop to 0.6 MPa and 0.5 m^3 at the end of the process. Determine (a) the amount of refrigerant that has escaped and (b) the final temperature of the refrigerant.



Analysis We take the tank as the system, which is a control volume since mass crosses the boundary. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances for this uniform-flow system can be expressed as

Mass balance: $m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_e = m_1 - m_2$

Energy balance:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

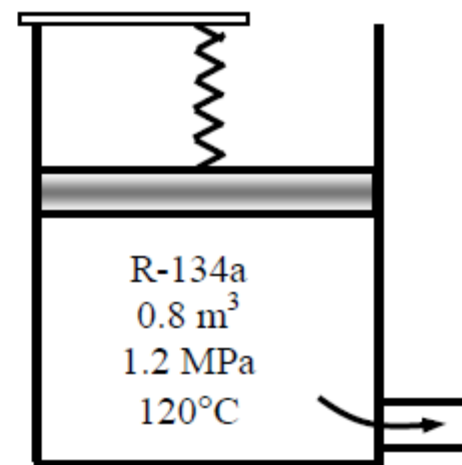
$$W_{b,\text{in}} - m_e h_e = m_2 u_2 - m_1 u_1 \quad (\text{since } Q \cong ke \cong pe \cong 0)$$

The initial mass and the relations for the final and exiting masses are

$$m_1 = \frac{V_1}{v_1} = \frac{0.8 \text{ m}^3}{0.02423 \text{ m}^3/\text{kg}} = 33.02 \text{ kg}$$

$$m_2 = \frac{V_2}{v_2} = \frac{0.5 \text{ m}^3}{v_2}$$

$$m_e = m_1 - m_2 = 33.02 - \frac{0.5 \text{ m}^3}{v_2}$$



Noting that the spring is linear, the boundary work can be determined from

$$W_{b,in} = \frac{P_1 + P_2}{2} (V_1 - V_2) = \frac{(1200 + 600) \text{ kPa}}{2} (0.8 - 0.5) \text{ m}^3 = 270 \text{ kJ}$$

Substituting the energy balance,

$$270 - \left(33.02 - \frac{0.5 \text{ m}^3}{v_2} \right) h_e = \left(\frac{0.5 \text{ m}^3}{v_2} \right) u_2 - (33.02 \text{ kg})(325.03 \text{ kJ/kg}) \quad (\text{Eq. 1})$$

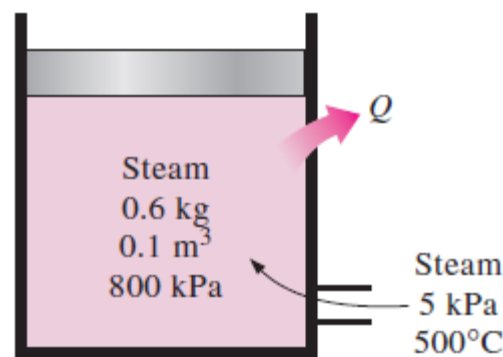
where the enthalpy of exiting fluid is assumed to be the average of initial and final enthalpies of the refrigerant in the cylinder. That is,

$$h_e = \frac{h_1 + h_2}{2} = \frac{(354.11 \text{ kJ/kg}) + h_2}{2}$$

Final state properties of the refrigerant (h_2 , u_2 , and v_2) are all functions of final pressure (known) and temperature (unknown). The solution may be obtained by a trial-error approach by trying different final state temperatures until Eq. (1) is satisfied. Or solving the above equations simultaneously using an equation solver with built-in thermodynamic functions such as EES, we obtain

$$T_2 = \mathbf{96.8^\circ C}, \quad m_e = \mathbf{22.47 \text{ kg}}, \quad h_2 = 336.20 \text{ kJ/kg}, \\ u_2 = 307.77 \text{ kJ/kg}, \quad v_2 = 0.04739 \text{ m}^3/\text{kg}, \quad m_2 = 10.55 \text{ kg}$$

5-144 A piston–cylinder device initially contains 0.6 kg of steam with a volume of 0.1 m³. The mass of the piston is such that it maintains a constant pressure of 800 kPa. The cylinder is connected through a valve to a supply line that carries steam at 5 MPa and 500°C. Now the valve is opened and steam is allowed to flow slowly into the cylinder until the volume of the cylinder doubles and the temperature in the cylinder reaches 250°C, at which point the valve is closed. Determine (a) the mass of steam that has entered and (b) the amount of heat transfer.



Properties The properties of steam at various states are (Tables A-4 through A-6)

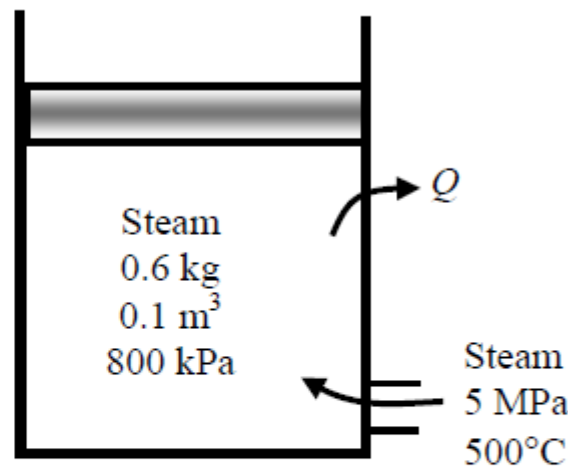
$$\nu_1 = \frac{V_1}{m_1} = \frac{0.1 \text{ m}^3}{0.6 \text{ kg}} = 0.16667 \text{ m}^3/\text{kg}$$

$$P_2 = P_1$$

$$\left. \begin{array}{l} P_1 = 800 \text{ kPa} \\ \nu_1 = 0.16667 \text{ m}^3/\text{kg} \end{array} \right\} u_1 = 2004.4 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_2 = 800 \text{ kPa} \\ T_2 = 250^\circ\text{C} \end{array} \right\} \begin{array}{l} \nu_2 = 0.29321 \text{ m}^3/\text{kg} \\ u_2 = 2715.9 \text{ kJ/kg} \end{array}$$

$$\left. \begin{array}{l} P_i = 5 \text{ MPa} \\ T_i = 500^\circ\text{C} \end{array} \right\} h_i = 3434.7 \text{ kJ/kg}$$



Analysis (a) We take the tank as the system, which is a control volume since mass crosses the boundary. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances for this uniform-flow system can be expressed as

Mass balance:

$$m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_i = m_2 - m_1$$

Energy balance:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} - W_{\text{b,out}} + m_i h_i = m_2 u_2 - m_1 u_1 \quad (\text{since } ke \cong pe \cong 0)$$

Noting that the pressure remains constant, the boundary work is determined from

$$W_{\text{b,out}} = P(\mathcal{V}_2 - \mathcal{V}_1) = (800 \text{ kPa})(2 \times 0.1 - 0.1) \text{ m}^3 = 80 \text{ kJ}$$

The final mass and the mass that has entered are

$$m_2 = \frac{\mathcal{V}_2}{\nu_2} = \frac{0.2 \text{ m}^3}{0.29321 \text{ m}^3/\text{kg}} = 0.682 \text{ kg}$$

$$m_i = m_2 - m_1 = 0.682 - 0.6 = \mathbf{0.082 \text{ kg}}$$

(b) Finally, substituting into energy balance equation

$$Q_{\text{in}} - 80 \text{ kJ} + (0.082 \text{ kg})(3434.7 \text{ kJ/kg}) = (0.682 \text{ kg})(2715.9 \text{ kJ/kg}) - (0.6 \text{ kg})(2004.4 \text{ kJ/kg})$$
$$Q_{\text{in}} = \mathbf{447.9 \text{ kJ}}$$

Review Problems

5-184 An adiabatic air compressor is to be powered by a direct-coupled adiabatic steam turbine that is also driving a generator. Steam enters the turbine at 12.5 MPa and 500°C at a rate of 25 kg/s and exits at 10 kPa and a quality of 0.92. Air enters the compressor at 98 kPa and 295 K at a rate of 10 kg/s and exits at 1 MPa and 620 K. Determine the net power delivered to the generator by the turbine.

Properties From the steam tables (Tables A-4 through 6)

$$\left. \begin{array}{l} P_3 = 12.5 \text{ MPa} \\ T_3 = 500^\circ\text{C} \end{array} \right\} h_3 = 3343.6 \text{ kJ/kg}$$

and

$$\left. \begin{array}{l} P_4 = 10 \text{ kPa} \\ x_4 = 0.92 \end{array} \right\} h_4 = h_f + x_4 h_{fg} = 191.81 + (0.92)(2392.1) = 2392.5 \text{ kJ/kg}$$

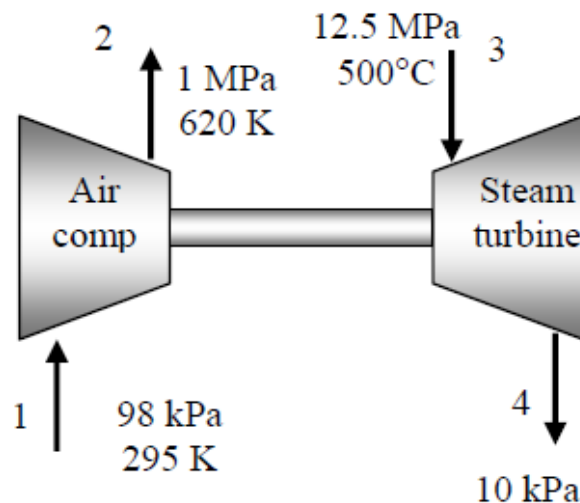
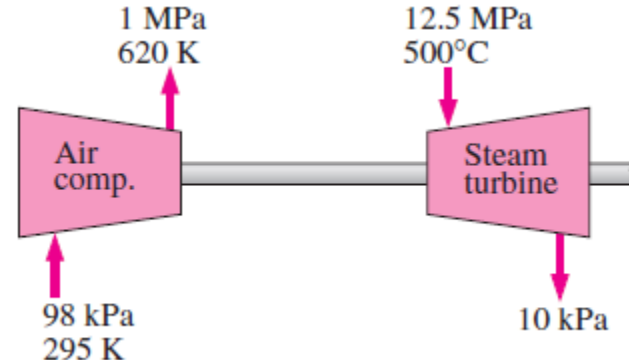
From the air table (Table A-17),

$$T_1 = 295 \text{ K} \longrightarrow h_1 = 295.17 \text{ kJ/kg}$$

$$T_2 = 620 \text{ K} \longrightarrow h_2 = 628.07 \text{ kJ/kg}$$

Analysis There is only one inlet and one exit for either device, and thus $\dot{m}_{\text{in}} = \dot{m}_{\text{out}} = \dot{m}$. We take either the turbine or the compressor as the system, which is a control volume since mass crosses the boundary. The energy balance for either steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\dot{\Delta E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$



$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

For the turbine and the compressor it becomes

Compressor:

$$\dot{W}_{\text{comp, in}} + \dot{m}_{\text{air}} h_1 = \dot{m}_{\text{air}} h_2 \rightarrow \dot{W}_{\text{comp, in}} = \dot{m}_{\text{air}} (h_2 - h_1)$$

Turbine:

$$\dot{m}_{\text{steam}} h_3 = \dot{W}_{\text{turb, out}} + \dot{m}_{\text{steam}} h_4 \rightarrow \dot{W}_{\text{turb, out}} = \dot{m}_{\text{steam}} (h_3 - h_4)$$

Substituting,

$$\dot{W}_{\text{comp, in}} = (10 \text{ kg/s})(628.07 - 295.17) \text{ kJ/kg} = 3329 \text{ kW}$$

$$\dot{W}_{\text{turb, out}} = (25 \text{ kg/s})(3343.6 - 2392.5) \text{ kJ/kg} = 23,777 \text{ kW}$$

Therefore,

$$\dot{W}_{\text{net, out}} = \dot{W}_{\text{turb, out}} - \dot{W}_{\text{comp, in}} = 23,777 - 3329 = \mathbf{20,448 \text{ kW}}$$

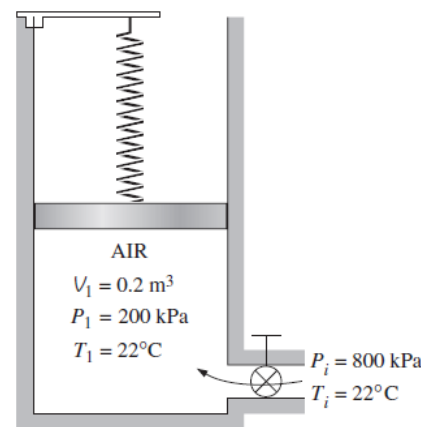
5-187



Steam enters a turbine steadily at 10 MPa and 550°C with a velocity of 60 m/s and leaves at 25 kPa with a quality of 95 percent. A heat loss of 30 kJ/kg occurs during the process. The inlet area of the turbine is 150 cm², and the exit area is 1400 cm². Determine (a) the mass flow rate of the steam, (b) the exit velocity, and (c) the power output.



An insulated vertical piston–cylinder device initially contains 0.2 m^3 of air at 200 kPa and 22°C . At this state, a linear spring touches the piston but exerts no force on it. The cylinder is connected by a valve to a line that supplies air at 800 kPa and 22°C . The valve is opened, and air from the high-pressure line is allowed to enter the cylinder. The valve is turned off when the pressure inside the cylinder reaches 600 kPa . If the enclosed volume inside the cylinder doubles during this process, determine (a) the mass of air that entered the cylinder, and (b) the final temperature of the air inside the cylinder.



Properties The gas constant of air is $R = 0.287 \text{ kJ/kg}\cdot\text{K}$ (Table A-1). The specific heats of air at room temperature are $c_v = 0.718$ and $c_p = 1.005 \text{ kJ/kg}\cdot\text{K}$ (Table A-2a). Also, $u = c_v T$ and $h = c_p T$.

Analysis We take the cylinder as the system, which is a control volume since mass crosses the boundary. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances for this uniform-flow system can be expressed as

Mass balance:

$$m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_i = m_2 - m_1$$

Energy balance:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

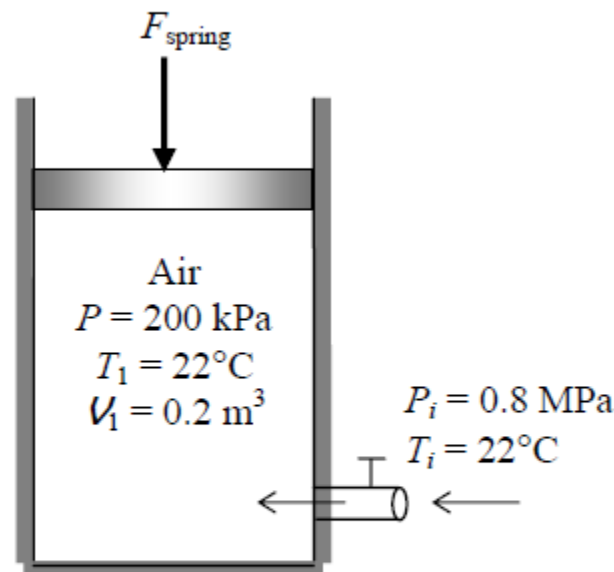
$$m_i h_i = W_{b,\text{out}} + m_2 u_2 - m_1 u_1 \quad (\text{since } Q \cong \text{ke} \cong \text{pe} \cong 0)$$

Combining the two relations,

$$(m_2 - m_1) h_i = W_{b,\text{out}} + m_2 u_2 - m_1 u_1$$

or,

$$(m_2 - m_1) c_p T_i = W_{b,\text{out}} + m_2 c_v T_2 - m_1 c_v T_1$$



The initial and the final masses in the tank are

$$m_1 = \frac{P_1 V_1}{RT_1} = \frac{(200 \text{ kPa})(0.2 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(295 \text{ K})} = 0.472 \text{ kg}$$
$$m_2 = \frac{P_2 V_2}{RT_2} = \frac{(600 \text{ kPa})(0.4 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})T_2} = \frac{836.2}{T_2}$$

Then from the mass balance becomes $m_i = m_2 - m_1 = \frac{836.2}{T_2} - 0.472$

The spring is a linear spring, and thus the boundary work for this process can be determined from

$$W_b = \text{Area} = \frac{P_1 + P_2}{2} (V_2 - V_1) = \frac{(200 + 600) \text{ kPa}}{2} (0.4 - 0.2) \text{ m}^3 = 80 \text{ kJ}$$

Substituting into the energy balance, the final temperature of air T_2 is determined to be

$$-80 = -\left(\frac{836.2}{T_2} - 0.472\right)(1.005)(295) + \left(\frac{836.2}{T_2}\right)(0.718)(T_2) - (0.472)(0.718)(295)$$

It yields $T_2 = \mathbf{344.1 \text{ K}}$

Thus, $m_2 = \frac{836.2}{T_2} = \frac{836.2}{344.1} = 2.430 \text{ kg}$

and $m_i = m_2 - m_1 = 2.430 - 0.472 = \mathbf{1.96 \text{ kg}}$

5-195 A piston–cylinder device initially contains 2 kg of refrigerant-134a at 800 kPa and 80°C. At this state, the piston is touching on a pair of stops at the top. The mass of the piston is such that a 500-kPa pressure is required to move it. A valve at the bottom of the tank is opened, and R-134a is withdrawn from the cylinder. After a while, the piston is observed to move and the valve is closed when half of the refrigerant is withdrawn from the tank and the temperature in the tank drops to 20°C. Determine (a) the work done and (b) the heat transfer. *Answers: (a) 11.6 kJ, (b) 60.7 kJ*

5-198 Steam enters a nozzle with a low velocity at 150°C and 200 kPa, and leaves as a saturated vapor at 75 kPa. There is a heat transfer from the nozzle to the surroundings in the amount of 26 kJ for every kilogram of steam flowing through the nozzle. Determine (a) the exit velocity of the steam and (b) the mass flow rate of the steam at the nozzle entrance if the nozzle exit area is 0.001 m².

Chapter 5

End of the Problems